

YAO-LUN YANG

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CONTACT INFORMATION

Star and Planet Formation Laboratory
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RESEARCH INTERESTS

Astrochemistry, Infall & Outflows, Early Stage Star Formation, Radiative Transfer Modeling, Atomic and Molecular Spectroscopy, and Infrared & Radio Astronomy.

PROFESSIONAL APPOINTMENTS

Mar. 2022 – present	Research Scientist (<i>indefinite term</i>) Star and Planet Formation Laboratory, RIKEN
Feb. 2022 – Jun. 2022	Visiting Scholar University of Virginia, USA
Feb. 2020 – Feb. 2022	Virginia Initiative on Cosmic Origins (VICO) Postdoctoral Fellow University of Virginia, USA
Aug. 2019 – Jan. 2020	Japan Society for the Promotion of Science (JSPS) Postdoctoral Fellow Host: Nami Sakai; RIKEN, Japan

EDUCATION

2019 Ph.D. Astronomy

The University of Texas at Austin, U.S.A. Advisors: Prof. Neal J. Evans II and Dr. Joel D. Green
Dissertation: *The Three-dimensional Structure and Kinematics of Protostellar Envelopes*

2015 M.A. Astronomy

The University of Texas at Austin, U.S.A. Advisor: Prof. Neal J. Evans II
Thesis: *The Class 0 Protostar BHR71: Herschel Observations and Dust Continuum Models*

2012 B.S. Physics

National Taiwan University, Taiwan Advisor: Dr. Ciska Kemper
Project: *Molecular Hydrogen in Diffuse Interstellar Medium of the Large Magellanic Cloud*

AWARDS, FUNDING (TOTAL: \$567,538 AS PI), AND RECOGNITIONS

RIKEN Incentive Research Project - Emergence of pre-biotic extraterrestrial world: Using machine-learning techniques to uncover the link of organic matter at the dawn of planetary systems
(¥2,700,000 as PI) 2026-2027

Grant-in-Aid for Scientific Research (A) - Tracing the Origins and Evolution of Interstellar Organic Molecules through Isotopic Observations: Toward Understanding the Primordial Materials of the Solar System (*translated*)

(total: ¥46,400,000; allocated: ¥2,200,000 as co-I) JSPS 2025-2028

Grant-in-Aid for Transformative Research Areas (A) - Exploring Chemical Evolution along Formation of Planetary-System by High Resolution/Sensitivity Observations

(total: ¥105,200,000; allocated: ¥1,200,000 as co-I, joined halfway) JSPS 2022-2024

Grant-in-Aid for Research Activity Start-up - Icy Origins of Organic Molecules Unveiled by JWST

(¥2,200,000 as PI) JSPS 2022-2023

JWST Cycle 1 GO Grant - (\$187,562 as PI)	STScI 2021
SOFIA Cycle 9 GO Grant - (\$134,000 as PI)	SOFIA 2021
Virginia Initiative on Cosmic Origins (VICO) Postdoctoral Fellowship	University of Virginia 2019
Japan Society for the Promotion of Science (JSPS) Postdoctoral Fellowship	Japan 2019
Concentration in Teaching and Mentoring	UT-Austin 2018
Professional Development Award (\$600)	UT-Austin 2018
University Graduate Continuing Fellowship (\$80,000)	UT-Austin 2017–2019
SOFIA Cycle 6 GO Grant - (\$41,000 as PI and \$39,000 as co-I)	SOFIA 2017
Fred T. Goetting, Jr. Memorial Endowed Presidential Fellowship (\$10,000)	UT-Austin 2016
Summer Internship (\$14,120)	STScI 2016
Outstanding Thesis Award (\$1,000)	School of Graduate Studies, UT-Austin 2016
SOFIA Cycle 4 GO Grant - (\$56,000 as PI and \$33,000 as co-I)	SOFIA 2015
Summer Student Fellowship (\$1,300)	ASIAA 2011
College Student Research Training Fellowship (\$1,500)	National Science Council, Taiwan 2011

PUBLICATIONS ([ORCID: 0000-0001-8227-2816](#))

First-Author and Significant Contribution Refereed Journal Articles

- [20] **Yang, Y.-L.**, Tan, J. C., Fedriani, R., & Zhang, Y. 2026, “Image Profile (IMPRO) Fitting of Massive Protostars. I. Method Development and Test Cases of Cepheus A and G35.20-0.74N”, submitted to *ApJ*, [arXiv:2508.12625](#)
- [19] Thompson, W. E., Bergner, J. E., Evans, N. J. II., **Yang, Y.-L.**, et al. 2026, “CORINOS. V. Radiative Transfer Effects in Protostellar Ice Observations”, *ApJ*, **1004**, 44
- [18] Oakey, P., **Yang, Y.-L.**, Tan, J. C., et al. 2026, “The SOMA Atomic Outflow Survey. I. An Atomic OI and Highly Ionized OIII Outflow from Massive Protostar G11.94-00.62”, *ApJ*, **1003**, 260
- [17] Turner, A., **Yang, Y.-L.**, Gross, R. E., et al. 2026, “A Chemistry-First Centered Icy Chemical Inventory of Protostellar Sources with JWST”, *ApJ*, **1000**, 17
- [16] Gross, R. E., **Yang, Y.-L.**, Cleaves, L. I., et al. 2026, “CORINOS IV: Baseline Fitting Effects on SO₂ Measurements in Class 0 Protostars with JWST MIRI”, *ApJ*, **998**, 68
- [15] Jiménez-Serra, I., Zeng, S., **Yang, Y.-L.**, et al. 2025, “Measuring metal sulfides in interstellar dust with PRIMA”, *JATIS*, **11**, 3
- [14] Hanawa, T., Okoda, Y., **Yang, Y.-L.**, Sakai, N. 2025, “Origin of the Shell Structure in the Primary Outflow from IRAS 15398–3359”, *ApJ*, **986**, 150
- [13] Okoda, Y., **Yang, Y.-L.**, Evans, N. J. II, et al. 2025, “CORINOS. III. Outflow Shocked Regions of the Low-mass Protostellar Source IRAS 15398–3359 with JWST and ALMA”, *ApJ*, **982**, 149
- [12] Salyk, C., **Yang, Y.-L.**, Pontoppidan, K. M., et al. 2024, “CORINOS II. JWST-MIRI detection of warm molecular gas from an embedded, disk-bearing protostar”, *ApJ*, **974**, 97
- [11] Pontoppidan, K. M., Evans, N. J. II, Bergner, J., & **Yang, Y.-L.** 2024, “A Constrained Dust Opacity for Models of Dense Clouds and Protostellar Envelopes”, *RNAAS*, **8**, 3
- [10] Zhang, Z. E., **Yang, Y.-L.**, Zhang, Y., et al. 2023, “The Perseus ALMA Chemistry Survey (PEACHES). II. Sulfur-bearing Species and Dust Polarization Revealing Shocked Regions in Protostars in the Perseus Molecular Cloud”, *ApJ*, **946**, 113
- [9] Evans, N. J. II, **Yang, Y.-L.**, Green, J. D., et al. 2023, “Models of Rotating Infall for the B335 Protostar”, *ApJ*, **943**, 90

- [8] **Yang, Y.-L.**, Green, J. D., Pontoppidan, K. M., et al. 2022, “CORINOS I: JWST/MIRI Spectroscopy and Imaging of a Class 0 protostar IRAS 15398–3359”, [ApJL, 941, L13](#)
- [7] **Yang, Y.-L.**, Evans, N. J. II, Karska, A., et al. 2022, “Atomic Shocks in the Outflow of L1551 IRS 5 Identified with SOFIA-upGREAT Observations of [O I]”, [ApJ, 925, 93](#)
- [6] **Yang, Y.-L.**, Sakai, N., Zhang, Y., et al. 2021, “The Perseus ALMA Chemistry Survey (PEACHES). I. The Complex Organic Molecules in Perseus Embedded Protostars”, [ApJ, 910, 20](#)
- [5] **Yang, Y.-L.**, Evans, N. J. II, Smith, A. et al. 2020, “Constraining the Infalling Envelope Models of Embedded Protostars: BHR71 and its Hot Corino”, [ApJ, 891, 1](#)
- [4] **Yang, Y.-L.**, Green, J. D., Evans, N. J. II, et al. 2018, “CO in Protostars (COPS): *Herschel*-SPIRE Spectroscopy of Embedded Protostars”, [ApJ, 860, 174](#)
- [3] **Yang, Y.-L.**, Evans, N. J. II, Green, J. D. et al. 2017, “The Class 0 Protostar BHR71: *Herschel* Observations and Dust Continuum Models”, [ApJ, 835, 259](#)
- [2] Green, J. D., **Yang, Y.-L.**, et al. 2016, “The CDF Archive: Herschel PACS and SPIRE Spectroscopic Data Pipeline and Products for Protostars and Young Stellar Objects”, [AJ, 151, 75](#)
- [1] Larson, R. L., Evans, N. J., Green, J. D., & **Yang, Y.-L.** 2015, “Evidence for Decay of Turbulence by MHD Shocks in the ISM via CO Emission”, [ApJ, 806, 70](#)

Other Refereed Journal Articles

- [57] Assani, K., Li, Z.-Y., Bergner, J. B., et al. 2026, “JWST Edge-on Disk Ice (JEDIce): Vibrationally hot, rotationally cold H₂ in the outer disk of Oph 163131 non-thermally excited by UV and cosmic rays”, accepted to AJ, [arXiv:2607.09407](#)
- [56] Chen, Y., van Dishoeck, E. F., Slavicinska, K., et al. 2026, “Probing the Ubiquity of Complex Ices in Protostars with JWST: The First Systematic Quantification of Weak Ice Bands between 6.8 and 7.9 μm ”, [ApJL, 1005, L30](#)
- [55] Lee, S., Lee, J.-L., Kim, C.-H., et al. 2026, “EPISODE II: Variability in the CO and H₂O rovibrational absorption lines in a periodically variable protostars EC 53”, [ApJ, 1005, 137](#)
- [54] Tychoniec, Ł., Francis, L., Navarro, M. G., et al. 2026, “JOYS: Launching and destruction of dust in protostellar jets. The case of BHR71-IRS1 with JWST/MIRI”, [A&A, 710, A370](#)
- [53] Kim, J., Lee, J.-E., Kim, C.-H., et al. 2026, “EPISODE IV: Ice Inventory in the Envelope of EC 53”, [ApJ, 1002, 202](#)
- [52] Bergner, J. B., Arulanantham, N., Dartois, E., et al. 2026, “JWST Edge-on Disk Ice (JEDIce): Program overview and ice survey results”, [ApJ, 1002, 1, 17](#)
- [51] Narang, M., Tyagi, H., Ohashi, N., et al. 2026, “Investigating the Nested Structure of the Outflow from the Low Luminosity Protostar IRAS 16253-2429 using JWST and ALMA”, [ApJ, 1000, 184](#)
- [50] Lee, S., Lee, J.-E., Kim, C.-H., et al. 2026, “EPISODE III: The Nested Jet/Outflow Morphology of EC 53 Revealed by JWST and ALMA”, [ApJ, 1000, 83](#)
- [49] Watson, D. M., Narang, M., Pittman, C. V., et al. 2026, “IPA. Accretion rate of a low-mass Class 0 protostar, measured via mid-infrared fluorescent OH emission”, [ApJ, 999, 264](#)
- [48] Federman, S., Megeath, S. T., Narang, M., et al. 2026, “The Structure and Kinematics of Three Class 0 Protostellar Jets from JWST”, [ApJ, 998, 282](#)
- [47] Feeney-Johansson, A., Aikawa, Y., Takakuwa, S., et al. 2026, “Early Planet Formation in Embedded Disks (eDisk). XIX. Structures of molecular outflows”, [AJ, 171, 172](#)

- [46] Dartois, E., Nobel, J. A., Bergner, J. B., **et al.** 2026, “The Flying Saucer edge-on disc’s Near Infrared silhouette revealed by the JWST JEDIce program”, [A&A, 705, L14](#)
- [45] Lee, J.-E., Kim, C.-H., Lee, S. **et al.** 2026, “Accretion bursts crystallize silicates in a planet-forming disk”, [Nature, 649, 853-858](#)
- [44] Sharma, R., Jørgensen, J. K., van’t Hoff, M. L. R., **et al.** 2025, “Early Planet Formation in Embedded Disks (eDisk). XX: Constraining the Chemical Tracers of Young Protostellar Sources”, [A&A, 704, A133](#)
- [43] Anderson, J. G., Bergner, J. B., **Yang, Y.-L.**, et al. 2025, “PEACHES IV: Tracing the Formation & Evolution of C₂H in Perseus Low-Mass Protostars”, [ApJ, 990, 150](#)
- [42] Zeng, S., J.-H. Jeong, Oyama, T., **et al.** 2025, “Determining the Methanol Deuteration in the Disk Around V883 Orionis with Laboratory Measured Spectroscopy”, [AJ, 170, 33](#)
- [41] Hsu, S.-Y., Lee, C.-F., Johnstone, D., **et al.** 2025, “ALMASOP. A Rotating Feature Rich in Complex Organic Molecules in a Protostellar Core”, [ApJ, 989, 56](#)
- [40] Telkamp, Z., Fedriani, R., Tan, J. C., **et al.** 2025, “The SOFIA Massive (SOMA) Star Formation Survey. V. Clustered Protostars”, [ApJ, 986, 15](#)
- [39] Brunken, N. G. C., Boogert, A. C. A., van Dishoeck, E. F., **et al.** 2025, “JWST observations of segregated ¹²CO₂ and ¹³CO₂ ices in protostellar envelopes”, [ACS Earth Space Chem. 2025, 9, 8, 1992–2003](#)
- [38] Kim, J., Lee, J.-L., Kim, C.-H., Jeong, W.-S., **Yang, Y.-L.** 2025, “Near- to mid-infrared spectroscopic study of ice analysis using the AKARI/IRC and Spitzer/IRS spectra”, [JKAS, 58, 111](#)
- [37] Karska, A., Figueira, M., Mirocha, A., **et al.** 2025, “SOFIA FIFI-LS spectroscopy of DR21 Main: energetics of the spatially-resolved outflow from a high-mass protostar”, [A&A, 697, A186](#)
- [36] Tyagi, H., Manoj, P., Narang, M., **et al.** 2025, “JWST-IPA: Chemical Inventory and Spatial Mapping of Ices in the Protostar HOPS370 – Evidence for an Opacity Hole and Thermal Processing of Ices”, [ApJ, 983, 110](#)
- [35] Crowe, S., Fedriani, R., Tan, J. C., **et al.** 2025, “The JWST-NIRCam View of Sagittarius C. I. Massive Star Formation and Protostellar Outflows”, [ApJ, 983, 19](#)
- [34] Lee, J.-E., Evans, N. J., Baek, G., **et al.** 2025, “A Natural Laboratory for Astrochemistry, a Variable Protostar B335”, [ApJL, 978, L3](#)
- [33] Taniguchi, K., Gorai, P., Tan, J. C., **et al.** 2024, “The SOFIA Massive (SOMA) Star Formation Q-band follow-up I. Carbon-chain chemistry of intermediate-mass protostars”, [A&A, 692, A65](#)
- [32] Hsu, S.-Y., Lee, C.-F., Liu, S.-Y., **et al.** 2024, “ALMASOP. The Localized and Chemically rich Features near the Bases of the Protostellar Jet in HOPS 87”, [ApJ 976, 29](#)
- [31] Rubinstein, A. E., Tyagi, H., Nazari, P., **et al.** 2024, “IPA. Class 0 Protostars Viewed in CO Emission Using JWST/NIRSpec”, [ApJ, 974, 112](#)
- [30] Gavino, S., Jørgensen, J. K., Sharma, R., **Yang, Y.-L.**, et al. 2024, “Early Planet Formation in Embedded Disks (eDisk). XI. A high-resolution view toward the BHR 71 Class 0 protostellar wide binary”, [ApJ, 974, 21](#)
- [29] Murillo, N. M., Fuchs, C. M., Harsono, D., **et al.** 2024, “The factors that influence protostellar multiplicity I: Gas temperature, density, and mass in Perseus with Nobeyama”, [A&A, 689, A267](#)
- [28] Slavicinska, K., van Dishoeck, E. F., Tychoniec, Ł, **et al.** 2024, “JWST detections of amorphous and crystalline HDO ice toward massive protostars”, [A&A, 688, A29](#)

- [27] Neufeld, D. A., Manoj, P., Tyagi, H., **et al.** 2024, “JWST/MIRI Detection of Suprathermal OH Rotational Emissions: Probing the Dissociation of the Water by Ly α Photons near the Protostar HOPS 370”, [ApJ](#), **966**, 2
- [26] Federman, S. A., Megeath, S. T., Rubinstein, A. E., **et al.** 2024, “Investigating Protostellar Accretion-driven Outflows across the Mass Spectrum: JWST NIRSpec Integral Field Unit 3-5 μ m Spectral Mapping of Five Young Protostars”, [ApJ](#), **966**, 1
- [25] Nazari, P., Rocha, W. R. M., Rubinstein, A. E., **et al.** 2024, “Hunt for complex cyanides in protostellar ices with JWST: Tentative detection of CH₃CN and C₂H₅CN”, [A&A](#), **686**, A71
- [24] Brunken, N. G. C., Rocha, W. R. M., van Dishoeck, E. F., **et al.** 2024, “JWST observations of ¹³CO₂ ice: Tracing the chemical environment and thermal history of ices in protostellar envelopes”, [A&A](#), **685**, 27
- [23] Narang, M., Manoj, P., Tyagi, H., **et al.** 2024, “Discovery of a Collimated Jet from the Low-luminosity Protostar IRAS 16253-2429 in a Quiescent Accretion Phase with the JWST”, [ApJL](#), **962**, L16
- [22] Hsu, S.-Y., Liu, S.-Y., Johnstone, D., **et al.** 2023, “ALMA Survey of Orion Planck Galactic Cold Clumps (ALMASOP): The Warm–Envelope Origin of Hot Corinos”, [ApJ](#), **956**, 120
- [21] Artur de la Villarmois, E., Guzmán, V. V., **Yang, Y.-L.**, **et al.** 2023, “The Perseus ALMA Chemical Survey (PEACHES). III. Sulfur-bearing species tracing accretion/ejection processes in young protostars”, [A&A](#), **678**, A124
- [20] Fedriani, R., Tan, J. C., Telkamp, Z., **et al.** 2023, “The SOFIA Massive (SOMA) Star Formation Survey. IV. Isolated Protostars”, [ApJ](#), **942**, 7
- [19] Law, C.-Y., Tan, J. C., Gorai, P., **et al.** 2022, “Isolated Massive Star Formation in G28.20–0.05”, [ApJ](#), **939**, 120
- [18] De Simone, M., Ceccarelli, C., Codella, C., **et al.** 2022, “Tracking the ice mantle history in the Solar-type Protostars of NGC 1333 IRAS 4”, [ApJL](#), **935**, L14
- [17] Zhang, Y., Tanaka K. E. I., Tan, J. C., **Yang, Y.-L.**, **et al.** 2022, “Massive Protostars in a Protocluster – A Multi-Scale ALMA View of G35.20-0.74N”, [ApJ](#), **936**, 68
- [16] Cheng, Y., Tobin, J. J., **Yang, Y.-L.**, **et al.** 2022, “Disks and Outflows in the Intermediate-mass Star Forming Region NGC 2071 IR”, [ApJ](#), **933**, 178
- [15] Bouvier, M., Ceccarelli, C., López-Sepulcre, A., Sakai, N., Yamamoto, S., & **Yang, Y.-L.** 2022, “The chemical nature of Orion protostars: Are ORANGES different from PEACHES? ORANGES II”, [ApJ](#), **929**, 10
- [14] van Gelder, M. L., Nazari, P., Tabone, B., **et al.** 2022, “Importance of source structure on complex organics emission. I. Observations of CH₃OH from low-mass to high-mass protostars”, [A&A](#), **A67**, 22
- [13] Hsu, S.-Y., Liu, S.-Y., Liu, T., **et al.** 2022, “ALMA Survey of Orion Planck Galactic Cold Clumps (ALMASOP): A Hot Corino Survey toward Protostellar Cores in the Orion Cloud”, [ApJ](#), **927**, 218
- [12] Liu, H. B., Tsai, A.-L., Chen, W. P., **et al.** 2021, “Millimeter-sized Dust Grains Appear Surviving the Water-sublimating Temperature in the Inner 10 au of the FU Ori Disk”, [ApJ](#), **923**, 270
- [11] Yun, H.-S., Lee, J.-E., Evans, N. J. II, **et al.** 2021, “TIMES II: Investigating the Relation Between Turbulence and Star-forming Environments in Molecular Clouds”, [ApJ](#), **921**, 31

- [10] Bouvier, M., López-Sepulcre, A., Ceccarelli, C, **et al.** 2021, “ORion Alma New GEneration Survey (ORANGES) I. Dust continuum and free-free emission of OMC-2/3 filament protostars”, [A&A, 653, A117](#)
- [9] Yun, H.-S., Lee, J.-E., Choi, Y., **et al.** 2021, “TIMES I: A Systematic Observation in Multiple Molecular Lines Toward the Orion A and Ophiuchus Clouds”, [ApJS, 256, 16](#)
- [8] Liu, M., Tan, J. C., De Buizer, J. M., **et al.** 2020, “The SOFIA Massive (SOMA) Star Formation Survey. III. From Intermediate- to High-Mass Protostars”, [ApJ, 904, 75](#)
- [7] Hsu, S.-Y., Liu, S.-Y., Liu, T., **et al.** 2020, “ALMA Survey of Orion Planck Galactic Cold Clumps (ALMASOP) I. Detection of New Hot Corinos with ACA”, [ApJ, 898, 107](#)
- [6] Liu, H. B., Mérand, A, Green, J. D., Pérez, S., Hales, A. S., **Yang, Y.-L.**, et al. 2019, “Diagnosing 0.1-10 au Scale Morphology of the FU Ori Disk using ALMA and VLTI/GRAVITY”, [ApJ, 884, 97](#)
- [5] Yi, H.-W., Lee, J.-L., Liu, T, **et al.** 2018, “Planck Cold Clumps in the λ Orionis complex. II. Environmental effects on core formation”, [ApJS, 236, 2](#)
- [4] Karska, A, Kaufman, M. J., Kristensen, L. E., **et al.** 2018, “The Herschel-PACS Legacy of Low-mass Protostars: Far-IR Gas Properties and Their Origin in FUV-illuminated C-shocks”, [ApJS, 235, 30](#)
- [3] Liu, T, Kim, K.-T., Juvela, M, **et al.** 2018, “The TOP-SCOPE Survey of Planck Galactic Cold Clumps: Survey Overview and Results of an Exemplar Source, PGCC G26.53+0.17”, [ApJS, 234, 28](#)
- [2] Green, J. D., Jones, O. C., Keller, L. D., **et al.** 2016, “The Mid-infrared Evolution of the FU Orionis Disk”, [ApJ, 832, 4](#)
- [1] Naslim, N., Kemper, F., Madden, S. C., **et al.** 2015, “Molecular Hydrogen Emission in the Interstellar Medium of the Large Magellanic Cloud”, [MNRAS, 446, 2490—2504](#)

Non-refereed Research Articles

- [14] Battersby, C., Beuther, H., Ginsburg, A., **et al.** 2025, “How Stars Assemble their Mass: Uncovering the Primary Mode of Stellar Accretion with PRIMA”, [PRIMA General Observer Science Book Volume 2](#)
- [13] Zeng, S., Jiménez-Serra, I., **Yang, Y.-L.**, et al. 2025, “Hidden Sulfur: tracing interstellar metal-sulfides with PRIMA”, [PRIMA General Observer Science Book Volume 2](#)
- [12] **Yang, Y.-L.**, Garatti o Caratti, A., Gutermuth, R., et al. 2025, “Surveying Atomic Outflows in Protostars Across Mass Spectrum”, [PRIMA General Observer Science Book Volume 2](#)
- [11] Tachibana, S., **Yang, Y.-L.**, Takigawa, A., et al. 2025, “Refractory minerals in the outflow from the massive young stellar object Orion Source I - Toward understanding the formation of the first solids in the Solar System”, [PRIMA General Observer Science Book Volume 2](#)
- [10] Megeath, T., Gutermuth, R., Oliveira, S., **et al.** 2025, “A Baseline Spectroscopic Survey of Protostars within 500 pc: Accretion Driven Feedback in Evolving and Outbursting Protostars”, [PRIMA General Observer Science Book Volume 2](#)
- [9] Cataldi, G., Furuya, K., Notsu, S., & **Yang, Y.-L.** 2025, “PRIMA can measure the CO and water content of exocomets”, [PRIMA General Observer Science Book Volume 2](#)
- [8] Tazaki, R., Honda, Mitsuhiko, **Yang, Y.-L.**, & Sargent, B. 2025, “Probing dust porosity in protoplanetary disks with far-IR polarization observations ”, [PRIMA General Observer Science Book Volume 2](#)

- [7] Sakai, N & **Yang, Y.-L.**, 2021, “Chemical Diversity in Young Protoplanetary Disk”, [ngVLA-J Memo Series](#)
- [6] **Yang, Y.-L.**, Evans, N. J., Smith, A., et al. 2020, “Direct Infall Signatures and Complex Organic Molecules toward an Isolated Embedded Protostar BHR 71”, *Origins: From the Protosun to the First Steps of Life*, Proceedings of the International Astronomical Union, 345, 312. [doi:10.1017/S1743921319001571](#)
- [5] Gutermuth, R., Offner, S., Arce, H., **et al.** 2019, “Dense Cores, Stellar Feedback and the Origins of Clustered Star Formation”, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 467, [BAAS](#), **51**, 467
- [4] Green, J., **Yang, Y.-L.**, Megeath, T., et al. 2019, “Variability in the Assembly of Protostellar Systems”, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 372, [BAAS](#), **51**, 372
- [3] Tobin, J., Offner, S., Sheehan, P., **et al.** 2019, “Measuring Protostar Masses: The Key to Protostellar Evolution”, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 189, [BAAS](#), **51**, 189
- [2] Tobin, J., Kounkel, M., Offner, S., **et al.** 2019, “The Formation and Evolution of Multiple Star Systems”, *Astro2020: Decadal Survey on Astronomy and Astrophysics*, science white papers, no. 187, [BAAS](#), **51**, 187
- [1] **Yang, Y.-L.**, Evans, N. J., & Green, J. 2015, “The Structure of Class 0 Protostars: BHR71 in Herschel View”, [Frank N. Bash Symposium 2015 \(BASH2015\)](#), 30

SELECTED OBSERVING PROGRAMS (AS PI OR SIGNIFICANT CO-I)

James Webb Space Telescope (JWST)

- *Early evolution of ice and UV processing in Orion’s Integral Shaped Filament*
Cycle 5 (as PI, 21.3 hr NIRSpec/MOS)
- *HEFE: High Angular Resolution observations of Stellar Emergence in Filamentary Environments*
(PI: T. Megeath)
Cycle 3 Large Program (as co-PI, 175 hrs NIRCAM / NIRSpec IFU / MIRI MRS)
- *Mapping the distribution of ices in the host environments of protostellar hot corinos* (PI: J. Jørgensen)
Cycle 3 Medium Program (as co-I, 67.5 hrs NIRCAM / NIRSpec MOS)
- *The JWST View of Icy Volatiles In Disks* (PI: J. Bergner)
Cycle 3 Medium Program (as co-I, 39.4 hrs NIRSpec IFU / MIRI MRS)
- *Cask-strength clouds: high percentage of methanol and HDO ices* (PI: M. Drozdovskaya)
Cycle 2 (as co-I, 8.7 hrs NIRCAM / NIRSpec MOS)
- *EPISODE: EC 53, the only known periodically variable infant star to chase the outburst in the next dynamical event* (PI: J.-E. Lee)
Cycle 2 (as co-I, 6.4 hrs MIRI MRS and NIRSpec IFU)
- *Blazing the Trails of COMs Ices to Gas*
Cycle 1 (as PI, 24.6 hrs MIRI MRS)
- *Investigating Protostellar Accretion Across the Mass Spectrum* (PI: T. Megeath)
Cycle 1 Medium Program (as co-I, 65.5 hrs MIRI MRS and NIRSpec IFU)

Acatama Large Millimeter/submillimeter Array (ALMA)

- *Toward a complete census of methanol budget in isolated protostars*
Cycle 9 & 10 (as PI, 17.8/16.9/44.0 hrs of 12m/ACA/TP)
- *Testing the origin of warm carbon-chain chemistry in Perseus protostars*
Cycle 10 (as PI, 54.5 hrs of ACA)
- *A hybrid approach to measure the 3D infall kinematics in an isolated protostellar core*
Cycle 10 (as PI, 4.8/5.5 hrs of 12m/ACA)
- *COMPASS: Complex Organic Molecules in Protostars with ALMA Spectral Surveys*
Cycle 9 Large Program (as co-I, PI: Jes Jørgensen, 125 hrs)
- *Directly measuring the progression of infall from the envelope to the disk-forming region of BHR 71*
Cycle 7 & 8 (as PI, 18.7 hrs)
- *Direct Detection of Infall in a Protostellar System*
Cycle 4 (as PI, 0.9 hr)
- *Spatiochemically Profiling the Inner Coma of C/2021 A1 (Leonard)*
Cycle 8 (as co-I, PI: Maria Drozdovskaya, 45.7 hrs)

Very Large Array (VLA)

- *Do Twins Always Look the Same?: Testing the Chemical Difference in a Protobinary*
2021 A (as PI, 15 hrs)

Stratospheric Observatory for Infrared Astronomy (SOFIA)

- *Probing the Radiative Cooling from Shocks and PDRs in Intermediate- and High-mass Protostars*
Cycle 9 (as PI, 23.1 hrs FIFI-LS)
- *Exploring Protostellar Winds with [OI]: Constraining models of shocked gas and PDR using L1551-IRS5*
Cycle 4 & 6 (as PI, 5.3 hrs & 4.1 hrs GREAT)
- *The Evolution of FU Orionis Disks*
Cycle 4 & 6 (as co-I, PI: Joel Green, 3 hrs & 3.9 hrs FORCAST)

Submillimeter Telescope (SMT)

- *Surveying the Chemical Diversity toward Taurus Embedded Protostars*
2022A (as PI, 40 hrs) & 2020B (as PI, 60 hrs)

Gemini-South

- *High Velocity Jets in the Outflow of BHR 71*
2020A (as co-PI, 6 hrs IGRINS)

APEX/FLASH⁺, 7.5 hrs (as PI)

Harlan J. Smith Telescope/DIAFI, 4 nights (as on-site observer)

IRTF/TEXES, 2014–2016, 4 nights (as co-I & on-site observer)

INVITED TALKS

ngVLA-J SWG Joint Meeting FY2025,	NAOJ, Tokyo, Japan 2026
Serminar at STARPLAN,	Univ. of Copenhagen, Denmark 2025
EA-ALMA Science Workshop 2025,	Kagoshima, Japan 2025
Physical and Chemical Processes of Astrophysical Interest: Astrochemistry at the JWST Era,	St. Florent, France 2025
LST Seminar,	Japan (<i>virtual</i>) 2025
PRL Workshop - Astrophysical Dust Ices: Insights from Recent Telescopes,	India (remote) 2025
COSPAR 2024,	Busan, Korea 2024
Astrophysics: The James Webb Space Telescope: from first light to new world views,	The Pontifical Academy of Sciences, Vatican City 2024
ngVLA-J Science Working Group meeting,	NAOJ, Toyko, Japan 2023
Japan Geoscience Union Meeting 2023,	Chiba, Japan 2023
EA-ALMA Science Workshop 2023,	New Taipei City, Taiwan 2023
Colloquium,	ASIAA, Taipei, Taiwan 2023
Dust, Ice, and Gas (DIG) Astrochemistry Meeting,	IASES (<i>virtual</i>), India/Japan 2022
Colloquium,	Chalmers University of Technology (<i>virtual</i>), Sweden 2022
SOFIA tele-talk,	SOFIA (<i>virtual</i>), USA 2022
SMA Seminar,	SAO/CfA (<i>virtual</i>), USA 2021
Astrochemistry Seminar,	NAOJ (<i>virtual</i>), Japan 2020
Colloquium,	NTHU, Hsinchu, Taiwan 2020
Seminar at Center of Astronomy and Gravitation,	NTNU, Taipei, Taiwan 2020
Colloquium,	ASIAA, Taipei, Taiwan 2020
ZUNA talk,	NRAO, VA, USA 2020
APEC Seminar,	IPMU, Chiba, Japan 2019
Star Formation Mini Symposium,	Kyung Hee University, Suwan, South Korea 2019
Colloquium,	NTHU, Taiwan 2019
Review talk, From Star to Planet Formation II,	Göteborg, Sweden 2019
CAS seminar,	Center for Astrochemical Studies, MPE, Germany 2019

SELECTED CONTRIBUTED TALKS

IAU-Kavli Symposium: Astrochemistry VII,	Traverse City, Michigan, USA 2023
First Science Results of JWST,	STScI, Baltimore, USA 2022
Next Generation Astrochemistry,	University of Tokyo, Tokyo, Japan 2022
Molecules in Extreme Environments: Near and Far,	NAOJ, Tokyo, Japan 2022
From Clouds to Planets II: The Astrochemical Link,	Berlin, Germany 2022
240 th AAS Meeting,	Pasadena (<i>hybrid</i>), CA 2022
NBIA Workshop on Radiation Transfer in Astrophysics,	Copenhagen (<i>hybrid</i>), Denmark 2022
Workshop on Interstellar Matter 2021,	Hokkaido (<i>hybrid</i>), Japan 2021
Astrochemistry in the JWST Era,	Leeds (<i>virtual</i>), UK 2021
2021 COSPAR,	virtual space, Earth 2021
Astrochemical Frontier,	virtual space, Earth 2020
2019 ALMA EA Development Workshop,	NAOJ, Tokyo, Japan 2019
Early Planet Formation in Embedded Disks,	University of Tokyo, Tokyo, Japan 2019
2018 ALMA EA Science/Development Workshop,	Osaka Pref. University, Osaka, Japan 2018
TUNA Talk,	NRAO/UVa, VA 2018

6th GMT Science Meeting: Stars Birth & Death,	Honolulu, HI 2018
72nd International Symposium on Molecular Spectroscopy,	UIUC, IL 2017
Star Formation 2016, Splinter session,	Exeter, UK 2016
Workshop on Dense Cores,	Monterey, CA 2014
Seminars at UT-Austin, NRAO/UVa, ASIAA, Subaru Telescope, IfA/U of Hawaii, Leiden Observa-	
tory, ESO-Garching, MPIA, STScI, East Asia Observatory, CfA/Harvard, U of Arizona, Kyung Hee	
University, Osaka University, and Osaka Prefecture University	

STUDENTS MENTORING

- Philip (Teddy) Oakey (undergraduate, UVA, 2023-present, co-advised with Prof. Jonathan Tan):
Lead the analysis of the SOMA/FIFI-LS survey to investigate the feedback of massive protostars from emission of shocks and PDRs.
- Jenny Margot Ramos Lázaro (CASSUM summer student, now master student at University of Chile, 2021–present):
Characterize the carbon-chain chemistry in Taurus embedded protostars using SMT observations.
- Taichi Hasegawa (undergraduate, U of Tokyo, 2025):
Analyze the emission of CO and H₂O in the JWST MIRI MRS spectrum of L483 using slab models.
- Moe Mishima (undergraduate, U of Tokyo, 2024)
Analyze the outflow and jet signatures from the H₂, [Fe II], and [Ne II] lines using JWST MIRI MRS observations in the CORINOS program.
- Lianis Reyes Rosa (graduate student, UVA, 2021–2023, co-advised with Prof. Jonathan Tan):
Investigate the feedback of massive protostars from emission of shocks and PDRs.
- Neha Bagalkot (undergraduate, UVA, 2021–2021, co-advised with Prof. Zhi-Yun Li):
Continuum and line radiative transfer calculations of star-forming cores.
- Pichaya Tositrakul (undergraduate, UVA, 2021):
Characterize the impact of dust optical depth to measurements of molecular abundance.
- Eva Greco (CASSUM summer student, UVA, 2020):
Study ALMA observations of CO outflows associated with a clustered massive star-forming regions, involving ALMA imaging and spectral analyses.
- Alyssa Ramos (undergraduate, UT-Austin, 2018):
Exploratory study on the complex organic molecules at the early phase of star formation, involving an archival study using the ALMA archive and simulating synthetic spectra of COMs.
- Rebecca Larson (undergraduate, UT-Austin, 2014–2016):
Constrain the decay of turbulence shocks with *Herschel* observations of starless molecular clouds. This project results in a publication on ApJ.

SERVICE AND PROFESSIONAL SOCIETY MEMBERSHIP

- Star and Planetary Science Lead for PRIMA-J team
- Co-lead of the ISM and Star Formation Working Group for PRIMA
- Founder and co-organizer of the “Astrochemistry Get-together Workshop” serving the astrochemistry community in the greater Tokyo area twice a year (5th workshop in Jun. 2026)
- SOC of a Lorentz Workshop, [Protostellar accretion and ejection with JWST and beyond](#)
- SOC of Session F3.4 in COSPAR 2024
- Member of NRAO Science Review Panels (2022–2023)
- Referee for ApJ, ApJS, MNRAS, A&A, Life Sciences in Space Research, and Nature Astronomy

- Subject-matter expert reviewer in NASA peer reviews (2021×2, 2022)
- External expert reviewer for JWST, HST, and ALMA Large Program
- Member of the ISSI International Team “Provenances of our Solar System’s relics”
- Affiliate of the eDisk ALMA Large Program
- Organizer of astro-ph at RIKEN, 2023-present
- Member of the Astronomical Society of Japan, 2022–present
- Member of the American Astronomical Society, 2017–2022
- AAS Meeting Chambliss Judge, 2019

PRESS RELEASES

- [NASA Webb Finds Young Sun-Like Star Forging, Spewing Common Crystals \(Japanese\)](#), Jan. 2026
- [A Baby Star in Action: B335 Offers a Natural Laboratory for Astrochemistry](#), Dec. 2024
- [Space telescope probes chemistry around a newborn star](#), Mar. 2022
- CORINOS I paper by Seoul National University (*Korean*), Dec. 2022
- CORINOS I paper by RIKEN, (*Japanese*), Dec. 2022
- Nature Astronomy, “*JWST takes stock of the freezer*”, Nov. 2022
- New Scientist, “*JWST reveals young star that may be about to give birth to a planet*”, Aug. 2022
- UVA Today, “*UVA Astronomers Will Map the Unmapped in Outer Space*”, Apr. 2022
- UVA Arts & Sciences Magazine March-April 2022, “*Mapping the Unmapped in Outer Space*”, Mar. 2022
- UVA Today, “*Like Peas in a Pod: UVA Astronomer’s Survey of Young Stars Published*”, Apr. 2021
- RIKEN, “*Amounts of organic molecules in planetary systems differ from early on*”, Apr. 2021

OUTREACH ACTIVITIES

- Talk, “Peering into the Origin of Solar System with JWST”, Space Cafe Tokyo, Mar. 2023
- Talk, “*Snow-covered Baby Stars and NASA’s Next Generation Space Telescope*”, Science on Screen, The Gem Theater, Mar. 2022
- Talk, “*JWST and Snow-covered Baby Stars*”, Massanutten Regional Library, Nov. 2021
- Organizer & Presenter, Astronomy on Tap ATX, Austin, TX, 2016–2019
Monthly astronomy talk held in a local bar joined by more than 250 audience
- Guest Host & Presenter, Astronomy on Tap Taipei, Taipei, Taiwan, 2019, 2020
- Talk, “*How to Make A Star*”, Westcave Preserve, Jan. 2015

(Last update on July 21, 2026)